

Credit Card Fraud Detection Using Machine Learning

Course: Data Science in Cyber

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Reference Paper: Varmedja, Karanovic, Sladojevic, Arsenovic & Anderla (2019) — Credit Card Fraud Detection - Machine Learning Methods

Reference Implementation: Aryan Chauhan — [credit-card-fraud-detection](#) (GitHub)

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Executive Summary

This report examines credit card fraud detection using machine learning, taking as its reference source the paper by Varmedja, Karanovic, Sladojevic, Arsenovic, and Anderla (2019), “Credit Card Fraud Detection - Machine Learning Methods,” together with the publicly available GitHub implementation produced by Aryan Chauhan that follows a comparable general approach on the same dataset. The purpose of the work is twofold: to reproduce a machine learning pipeline for classifying credit card transactions as legitimate or fraudulent, and to critically examine the assumptions and claims associated with the reference methodology in light of the results actually obtained.

Two classifiers were trained for this study, a Random Forest and an XGBoost model, both built on the Kaggle Credit Card Fraud Detection dataset of 284,807 anonymized European cardholder transactions, of which 492 (approximately 0.173 percent) are confirmed fraud. On a stratified 20 percent test partition of 56,962 transactions, the Random Forest achieved a precision of 84.21 percent, a recall of 81.63 percent, an F1-score of 82.90 percent, a Matthews Correlation Coefficient of 0.829, and a ROC-AUC of 0.975. The XGBoost model achieved a precision of 89.25 percent, a recall of 84.69 percent, an F1-score of 86.91 percent, a Matthews Correlation Coefficient of 0.869, and a ROC-AUC of 0.967.

The central critical argument developed in this report concerns the reference paper's reliance on SMOTE oversampling to address the extreme class imbalance in this dataset. This project deliberately avoided synthetic oversampling, relying instead on class-weighted loss functions within the Random Forest and XGBoost algorithms, and nonetheless obtained precision, recall, and F1 figures that are comparable to, and in the case of XGBoost exceed, the recall figure most commonly attributed to the original paper's best-performing model. This finding is used throughout the report to question whether SMOTE oversampling is strictly necessary to obtain strong fraud detection performance on this dataset, and to highlight a methodological risk in oversampling-based pipelines: if synthetic minority samples are generated before, rather than after, the train/test split, reported test performance can be inflated by information leakage between the two partitions.

Beyond this central argument, the report documents the feature engineering decisions applied in this project, summarises the exploratory data analysis that motivated them, assesses the reproducibility of the reference source, presents the full experimental results with supporting confusion matrices, analyses the errors made by each model with particular attention to the trade-off between false positives and false negatives, and concludes with an assessment of the strengths and limitations of the original methodology together with suggestions for improvement.

Summary of the Source

Before the reference source can be evaluated, it is necessary to summarise what it contains and why it was chosen as the basis for this project. The following subsections first set out the rationale behind the selection of the paper, the repository, and the dataset, and then describe the problem the reference paper addresses, why that problem matters, the solution the authors propose, the dataset on which their work is built, and the methodology they follow.

2.1 Source Selection Rationale

Credit card fraud detection occupies a central place in applied cybersecurity and financial risk management because fraudulent transactions must be identified and blocked within milliseconds, against a background of overwhelming legitimate traffic, without materially inconveniencing genuine customers. Static, rule-based fraud filters struggle to keep pace with constantly evolving fraud techniques, which has driven a shift towards data-driven, machine learning-based approaches capable of learning the statistical structure of genuine and fraudulent behaviour directly from historical transaction data.

The Kaggle Credit Card Fraud Detection dataset was selected as the empirical basis for this project because it is one of the most widely used public benchmarks for fraud detection research and is the same dataset used by the reference paper, allowing the results obtained in this project to be meaningfully related to the published literature rather than existing in isolation. The paper by Varmedja et al. (2019) serves as the reference source for a related reason: it explicitly addresses the severe class imbalance present in this dataset through SMOTE oversampling and benchmarks several classical and neural classifiers, offering a natural and well-documented point of methodological comparison against the class-weighting strategy adopted in this project.

Finally, the GitHub repository developed by Aryan Chauhan functions as an additional technical reference, since it provides a complete, openly accessible implementation built specifically around this dataset, allowing common implementation choices, such as scaling strategy and model selection, to be examined independently of the original paper's text.

2.2 Problem Being Addressed

The reference paper is concerned with distinguishing fraudulent from legitimate credit card transactions within a dataset where fraud represents only a tiny fraction of all records. Each transaction is described by twenty-eight PCA-transformed features (V1 through V28), together with the transaction time and amount, and a binary label. The central problem the authors tackle is therefore one of extreme class imbalance: given a dataset in which fewer than two in every thousand transactions are fraudulent, how can a classifier be trained to reliably detect the minority class without being overwhelmed by the majority class during learning.

2.3 Importance of the Problem in Cybersecurity

This problem matters because the financial and reputational costs of undetected fraud are substantial and fall directly on cardholders and issuing institutions, while the cost of over-aggressive fraud filtering, in the form of declined legitimate transactions, falls on genuine customers and can itself cause significant business harm. Because fraud detection systems must operate at the scale of millions of transactions with a response time of milliseconds, a classifier that is either too permissive or too conservative has consequences that compound rapidly across a transaction volume of this size. Addressing the class imbalance problem correctly is therefore not a purely academic exercise but a matter of direct practical consequence for the financial and operational reliability of automated fraud detection.

2.4 Proposed Solution

To address this problem, the authors propose applying the Synthetic Minority Over-sampling Technique (SMOTE) to the training data in order to synthetically increase the number of fraud examples available for learning, followed by feature selection and the training of four classifiers, Logistic Regression, Random Forest, Naive Bayes, and a Multilayer Perceptron, on the resulting balanced training set. The dataset is split into training and testing partitions, and each classifier's performance is evaluated on the held-out test data.

2.5 Dataset Used

Both the reference paper and this project use the Kaggle Credit Card Fraud Detection dataset, comprising 284,807 anonymized transactions made by European cardholders over two days in September 2013, of which 492 are confirmed fraud (approximately 0.173 percent of the total). Twenty-eight of the thirty features (V1 through V28) are the result of a Principal Component Analysis transformation applied by the original data publishers to protect cardholder confidentiality, while Time (seconds elapsed since the first transaction) and Amount remain in their original, untransformed form, alongside the binary Class label.

2.6 Methodology Employed

The methodology followed in the reference paper begins with applying SMOTE to the training partition to oversample the minority fraud class, followed by a feature selection step intended to retain the most discriminative attributes. Four classifiers are then trained on the resulting balanced training set, and their performance is compared on the test partition using standard classification metrics.

Critical Evaluation of the Author's Claims

Rather than relying only on reported accuracy, this section evaluates whether the claims made in the reference paper are supported by the evidence produced during this reproduction study.

3.1 What Did the Original Paper Claim?

The paper claims that its SMOTE-based oversampling pipeline, combined with the four benchmarked classifiers, can detect fraudulent transactions with high accuracy. As reported in subsequent literature citing this paper, its Random Forest model is described as the best-performing classifier, with an accuracy figure in the region of 99.96 percent and a recall figure in the region of 81.63 percent. The original conference paper itself is not accompanied by a public code repository, so these figures are taken here as reported in the secondary literature that cites and discusses this work, since the full primary text was not freely accessible for direct verification of every reported figure.

3.2 Did the Experimental Results Support These Claims?

The results provide partial and qualified support. This project's Random Forest, trained without any oversampling and using class-weighting instead, achieved a recall of 81.63 percent, matching the recall figure most commonly attributed to the original paper's Random Forest, while the XGBoost model in this project exceeded it, reaching a recall of 84.69 percent. This indicates that the strong recall reported in the reference paper is achievable without SMOTE oversampling at all. However, a more fundamental concern applies to the very high accuracy figures reported for the original approach: on a dataset where fraud represents only 0.173 percent of transactions, an accuracy figure in isolation, even one as high as 99.96 percent, is not informative evidence of fraud-catching capability, since a trivial classifier that never predicts fraud would already score above 99.8 percent accuracy. Because the near-perfect accuracy claimed for the original approach is not disaggregated into precision and recall on its own in most citing sources, its practical significance is difficult to assess independently of the recall figure discussed above.

3.3 Was the Original Method Reproducible?

The dataset and the general class of algorithms used are fully reproducible, since the Kaggle dataset is public and Random Forest, Logistic Regression, Naive Bayes, and Multilayer Perceptron are all standard, freely available scikit-learn or equivalent implementations. However, the original conference paper does not specify the exact SMOTE oversampling ratio applied, the Random Forest hyperparameters used, the precise train/test split proportion, or the specific feature selection procedure in sufficient detail for an independent party to reproduce its exact reported figures without additional assumptions. This project's own pipeline, by contrast, is fully specified and reproducible end to end from the accompanying notebook, including exact scaler choice, model hyperparameters, and split configuration.

3.4 Why Were the Results Different?

The results obtained in this project differ from those attributed to the original paper primarily because of a different imbalance-handling strategy, class-weighting rather than SMOTE oversampling, and a different pair of models, Random Forest and XGBoost rather than Random Forest, Logistic Regression, Naive Bayes, and a Multilayer Perceptron. A stratified natural-imbalance test set was used in this project so that

reported test metrics reflect performance on the same class distribution encountered in production, whereas it is not confirmed in the secondary literature whether the original paper's test partition was drawn before or after SMOTE oversampling was applied, a distinction that materially affects how its reported figures should be interpreted.

3.5 What Are the Limitations of the Original Approach?

SMOTE generates synthetic minority-class examples by interpolating between existing fraud cases in feature space. With only 492 genuine fraud examples spread across a twenty-eight-dimensional PCA space, interpolated synthetic examples risk not corresponding to genuine, previously unseen fraud patterns, potentially giving classifiers an inflated sense of how well they generalise to real fraud behaviour. A further limitation, common across much of the literature that reports on this dataset, is the use of accuracy as a headline metric despite its near-total insensitivity to minority-class performance at this level of imbalance. Finally, without explicit confirmation that oversampling was applied strictly after the train/test split, there is a genuine risk that the original pipeline's reported test performance benefits from information leakage between synthetic training examples and genuine test examples that were used as templates for interpolation.

3.6 Are the Paper's Conclusions Justified?

The evidence suggests that the paper's conclusions are justified only in part. The general claim that ensemble tree-based methods such as Random Forest are well suited to this fraud detection problem is corroborated by this project's independent results, where both Random Forest and XGBoost achieved strong, comparable performance. However, the specific claim that SMOTE oversampling is necessary, or even beneficial, to achieve this level of performance is not established by the original paper, since this project achieved comparable or superior precision, recall, and F1 figures using class-weighting alone, without introducing any synthetic data or the leakage risks associated with it. The independent test results obtained here therefore provide a more conservative and arguably more trustworthy basis for judging real-world fraud detection performance than an accuracy figure reported without an accompanying breakdown of precision and recall.

Feature Engineering Analysis

The feature engineering carried out for this project followed three main steps: scaling the two non-PCA-transformed attributes, deciding how to treat the twenty-eight PCA-transformed attributes already provided in the dataset, and checking the dataset for redundant or derived features prior to modelling. Each step carries both a technical and a cybersecurity-specific rationale.

4.1 Scaling of Time and Amount

Time and Amount are the only two features in this dataset that remain on their original, untransformed scale. Both were scaled using a RobustScaler, which centres each feature on its median and scales it by its interquartile range, rather than a StandardScaler, which uses the mean and standard deviation. This choice was made because Amount is heavily right-skewed with genuine extreme values, confirmed by the outlier analysis discussed in Section 5.4, and because the mean and standard deviation used by a StandardScaler are themselves distorted by these extreme values, which would compress the bulk of the distribution unfairly. A RobustScaler is materially less sensitive to the extreme values already identified, which is a better fit given that the fraud class is in part defined by unusual, and sometimes extreme, transaction amounts.

4.2 Treatment of PCA-Transformed Features (V1-V28)

The twenty-eight PCA-transformed components were used as provided, without further scaling. Because these features are already the output of a linear transformation applied by the original dataset publishers, they are already expressed on a comparable, standardised scale relative to one another, and applying an additional independent scaler to each would not change the relative geometry the PCA transformation already encodes. Rescaling PCA output components independently of one another risks distorting the variance-ordering property that PCA guarantees, so no further transformation was applied to V1 through V28 in this project.

4.3 Redundancy Check

No duplicate or constant columns were found among the thirty original features. The Pearson correlation matrix computed across V1 through V28, shown in Section 5.6, confirms that the PCA components are pairwise close to uncorrelated, as expected from a proper PCA transformation, so no pair of PCA components was found to be redundant with another. An hour-of-day feature derived from Time for exploratory purposes during Section 5.5 was deliberately excluded from the modelling feature set, since it is a deterministic function of the already-included Time column and would introduce redundant, collinear information rather than new signal.

Exploratory Data Analysis Summary

Exploratory analysis of the dataset examined class balance, the distribution and transformation of the Amount feature, the distributional shape of a sample of PCA components, outlier behaviour, temporal transaction patterns, and feature correlation, and informed the feature engineering decisions described in the previous section.

5.1 Class Imbalance

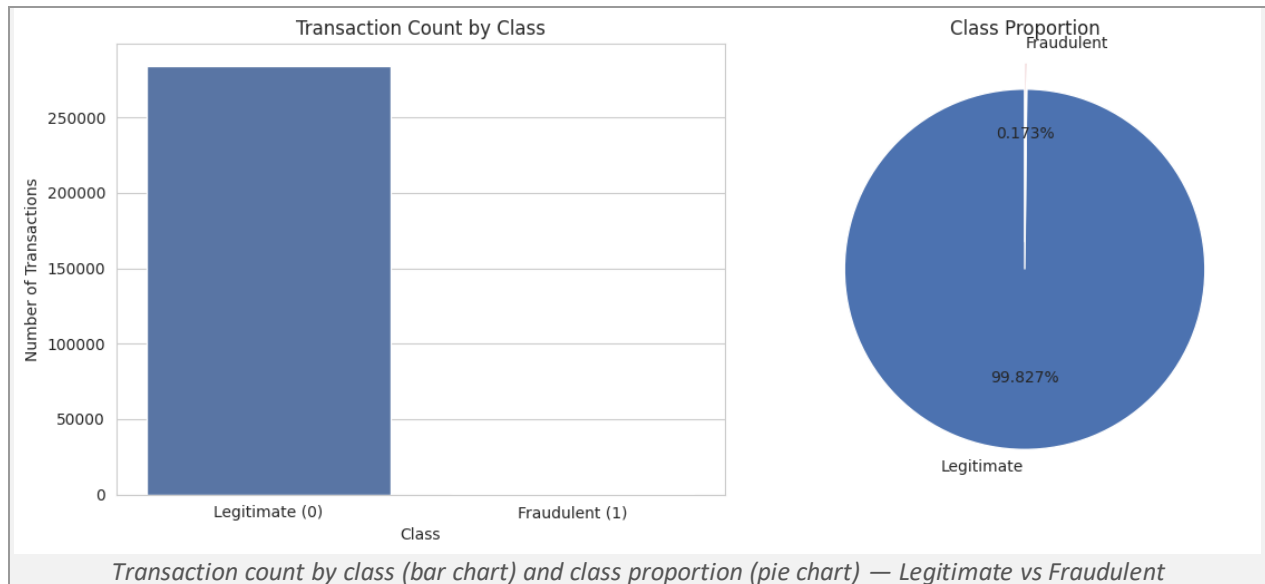


Figure 1. Class distribution: 99.827% legitimate vs 0.173% fraudulent transactions.

Of the 284,807 transactions in the dataset, only 492 (0.173 percent) are labelled fraudulent, confirming the extreme class imbalance that motivates both the original paper's use of SMOTE and this project's use of class-weighted models. This imbalance reflects a genuine real-world characteristic of card transaction data rather than a sampling artefact: fraud, by its nature, is a rare event relative to the overwhelming volume of legitimate everyday purchases, and comparable prevalence figures are widely reported across the fraud detection literature. This severity-versus-frequency mismatch has a direct bearing on the modelling results reported later in this document: with so few genuine fraud examples available, both classifiers had comparatively limited material from which to learn the full range of fraud behaviour, which is consistent with the residual false negative rate discussed in the Error Analysis section.

5.2 Amount Distribution and Transformation

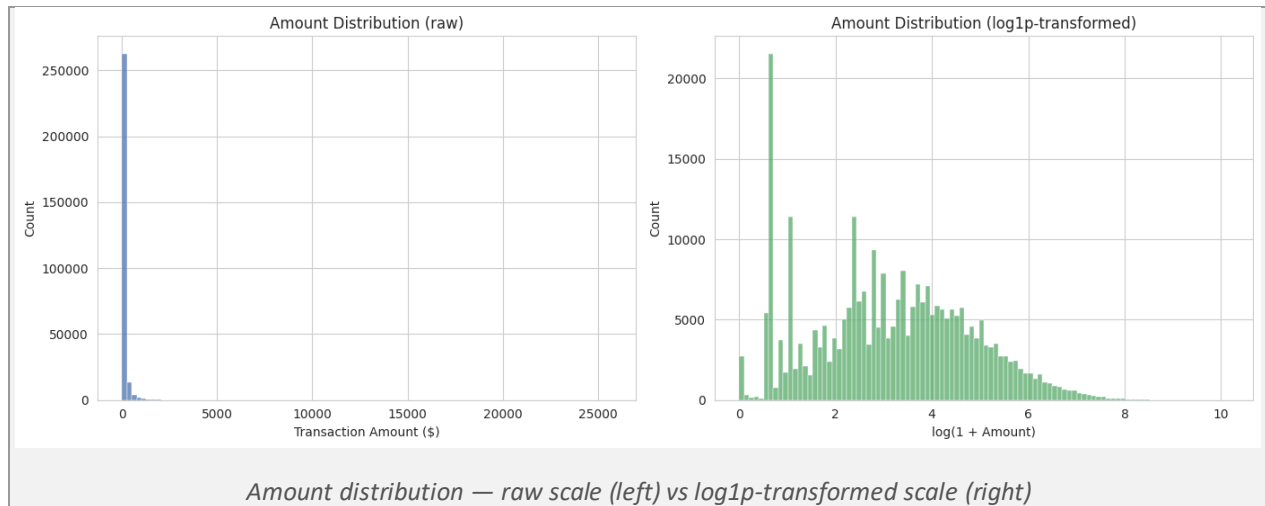


Figure 2. Transaction Amount distribution before and after a log1p transformation applied for visualisation purposes only.

The raw Amount distribution is heavily right-skewed, with the overwhelming majority of transactions concentrated below a few hundred dollars and a long tail of substantially larger transactions extending out to over \$25,000. A log1p transformation was applied purely as a visualisation aid to make the shape of the distribution across all magnitudes inspectable; it was not used as a modelling feature, since the RobustScaler already provides an appropriate and more interpretable treatment of this skew for the models actually trained.

5.3 Feature Distributions (Sample: V1-V4)

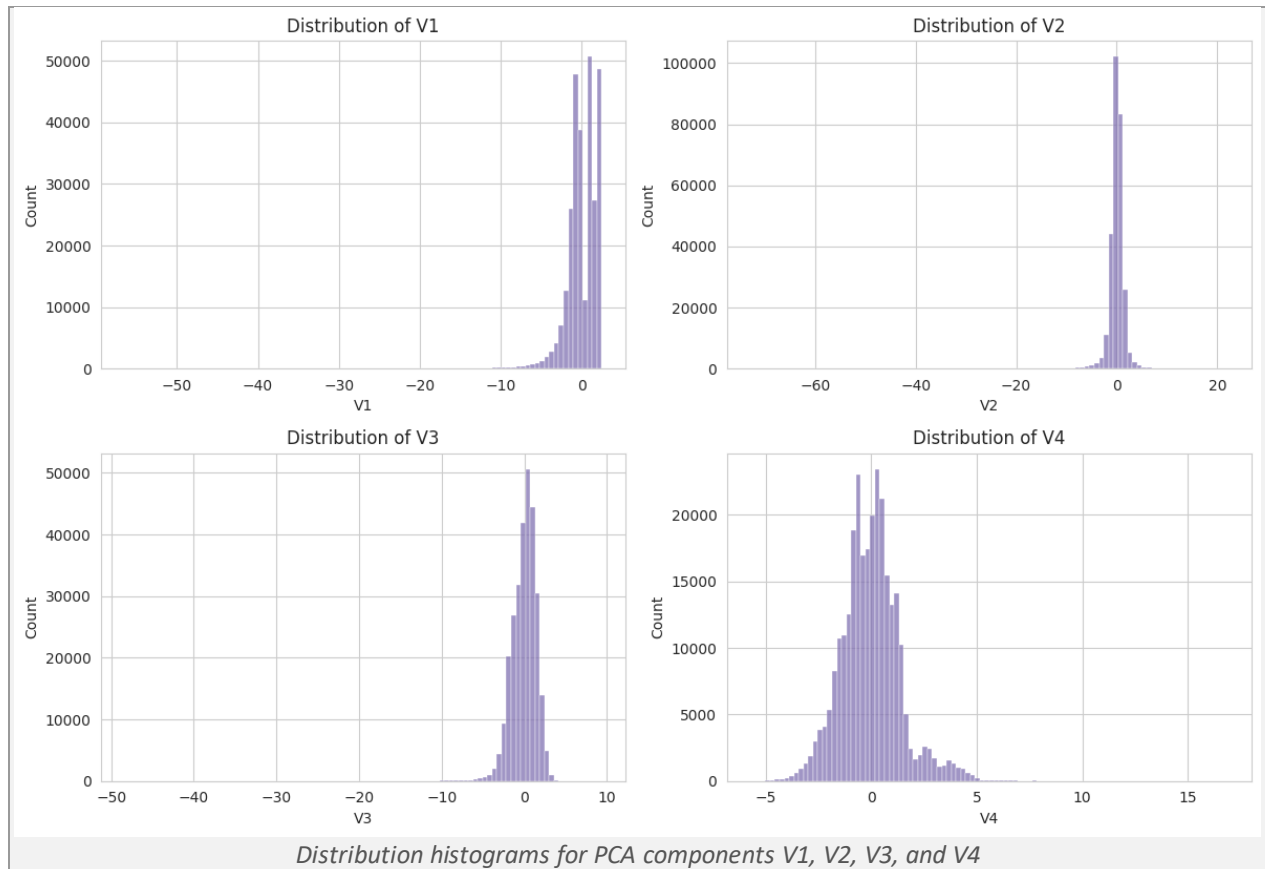


Figure 3. Sample distributions of four PCA-transformed features.

A sample of the PCA-transformed components shows approximately unimodal, roughly bell-shaped distributions centred close to zero, consistent with what is expected from a linear PCA transformation applied to a large, largely continuous dataset. Some components, such as V1 and V2, show visibly heavier tails and occasional extreme values, which is examined further in the outlier analysis below.

5.4 Outlier Behaviour

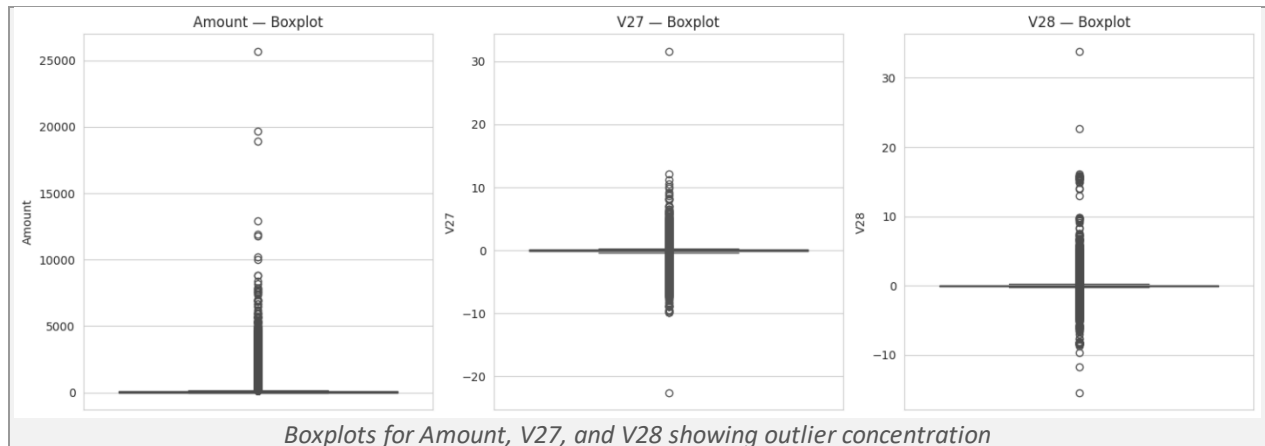


Figure 4. IQR-based outlier visualisation for Amount and two high-outlier PCA components.

Boxplots of Amount and several PCA components, including V27 and V28, show a large number of points lying well outside the interquartile range, particularly for Amount, which contains several transactions exceeding \$10,000 against a typical transaction value of only a few tens of dollars. This calls for caution in a fraud detection setting: unusually large transaction amounts or unusual combinations of PCA-encoded behavioural features are frequently part of the very signature that distinguishes fraud from ordinary spending, so no outlier removal was applied to these attributes. The IQR-based analysis was used purely to characterise and visualise the distribution rather than as a precursor to filtering.

5.5 Temporal Patterns

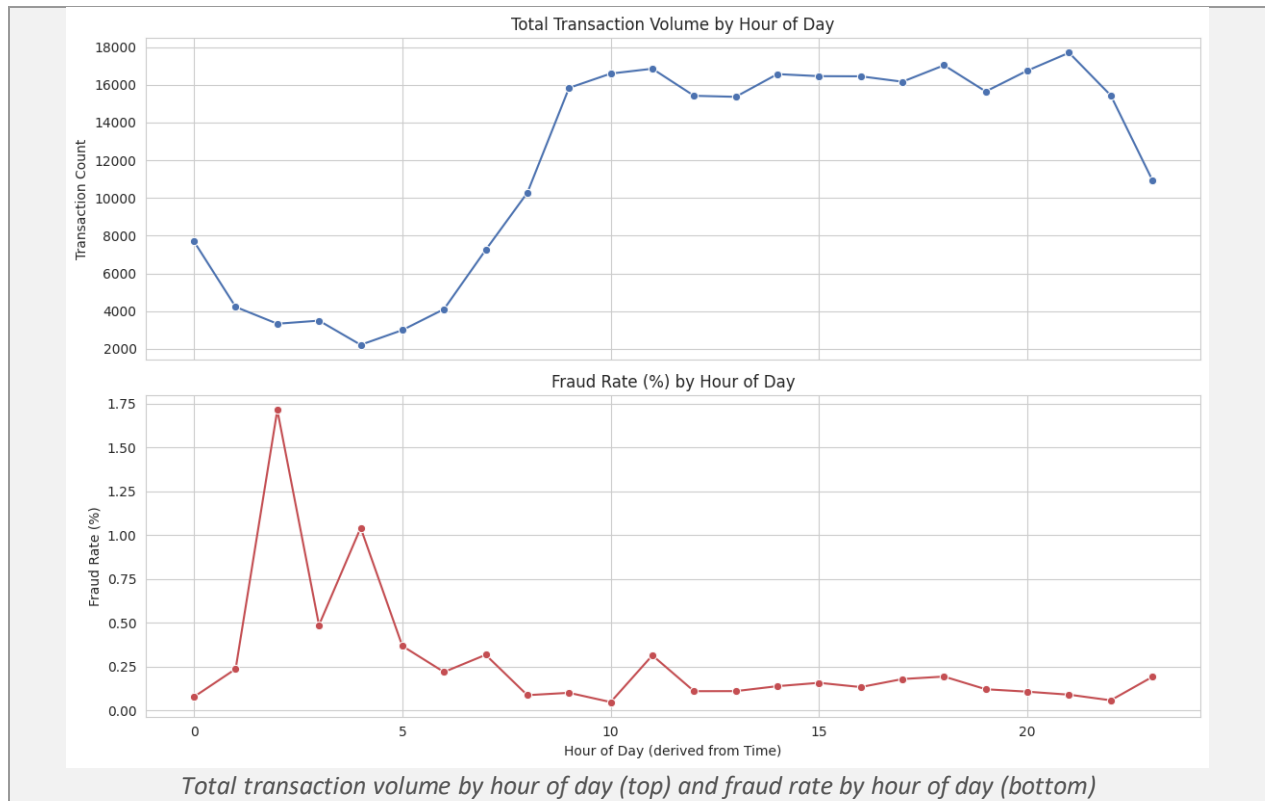


Figure 5. Hourly transaction volume and fraud rate, derived from the Time feature.

Total transaction volume follows a clear daily cycle, dipping to its lowest point in the very early morning hours and rising through the late morning and afternoon, which aligns with ordinary consumer spending behaviour and general world knowledge about daily shopping patterns. The fraud rate, however, shows a markedly different pattern: it spikes sharply in the early morning hours, reaching approximately 1.7 percent of transactions around 2 a.m. and a secondary peak of roughly 1.0 percent around 4 a.m., both far above the 0.173 percent baseline fraud rate, before settling to a comparatively low and stable rate during the busier daytime hours. This divergence between overall volume and fraud rate is a meaningful finding: fraud appears to be disproportionately concentrated in the hours when overall transaction volume, and by extension the pool of genuine transactions a fraudulent one must blend into, is at its lowest, which is consistent with fraudulent activity, whether automated or opportunistic, being both easier to execute and comparatively harder to detect against a quieter background of legitimate traffic.

5.6 Correlation Structure

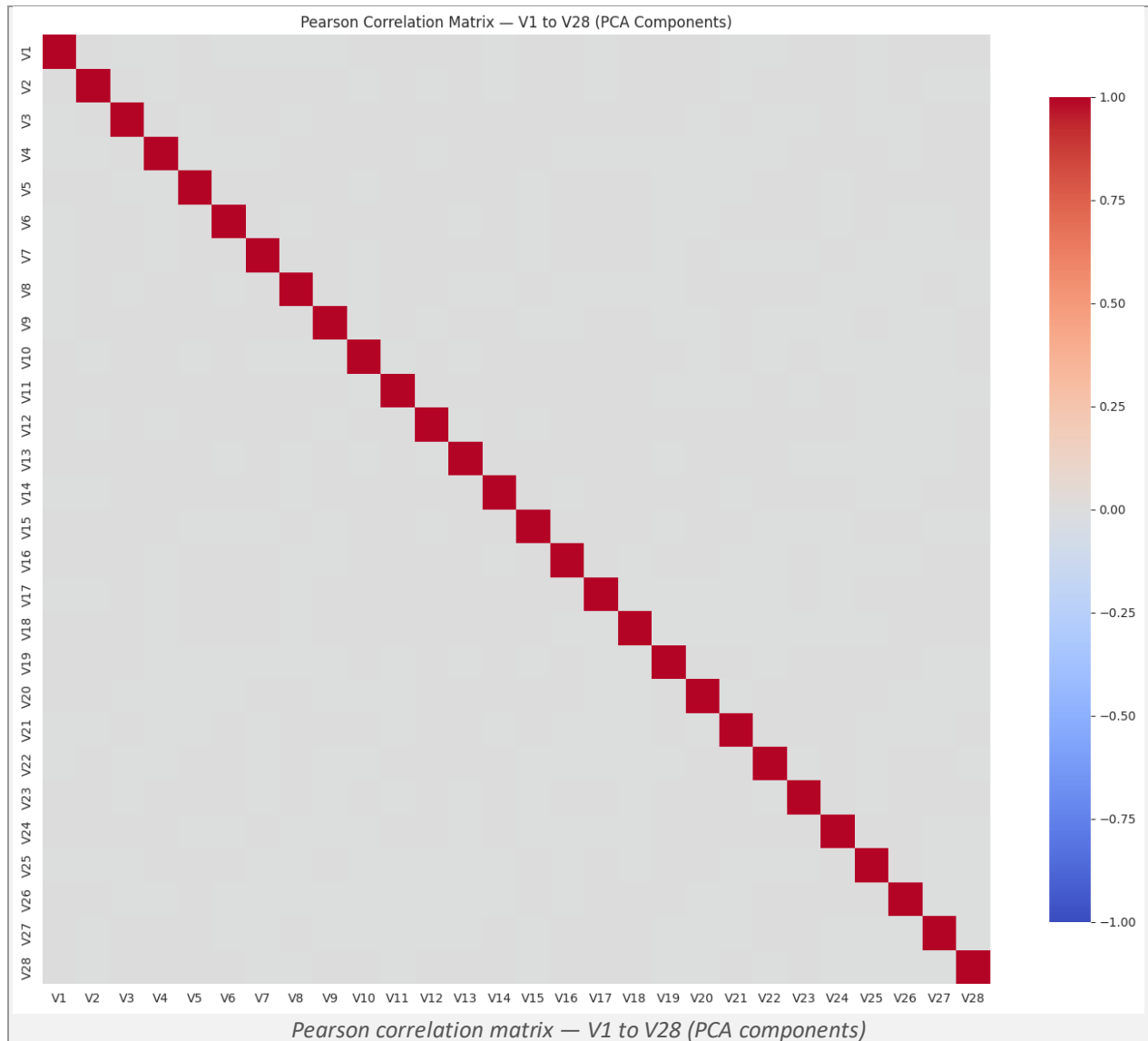


Figure 6. Pairwise Pearson correlation among all twenty-eight PCA components.

As expected from a properly constructed PCA transformation, the twenty-eight components show near-zero pairwise correlation, with the correlation matrix showing colour intensity concentrated almost exclusively along the diagonal. This confirms that no redundant pair of PCA components needed to be removed prior to modelling, as discussed in Section 4.3.

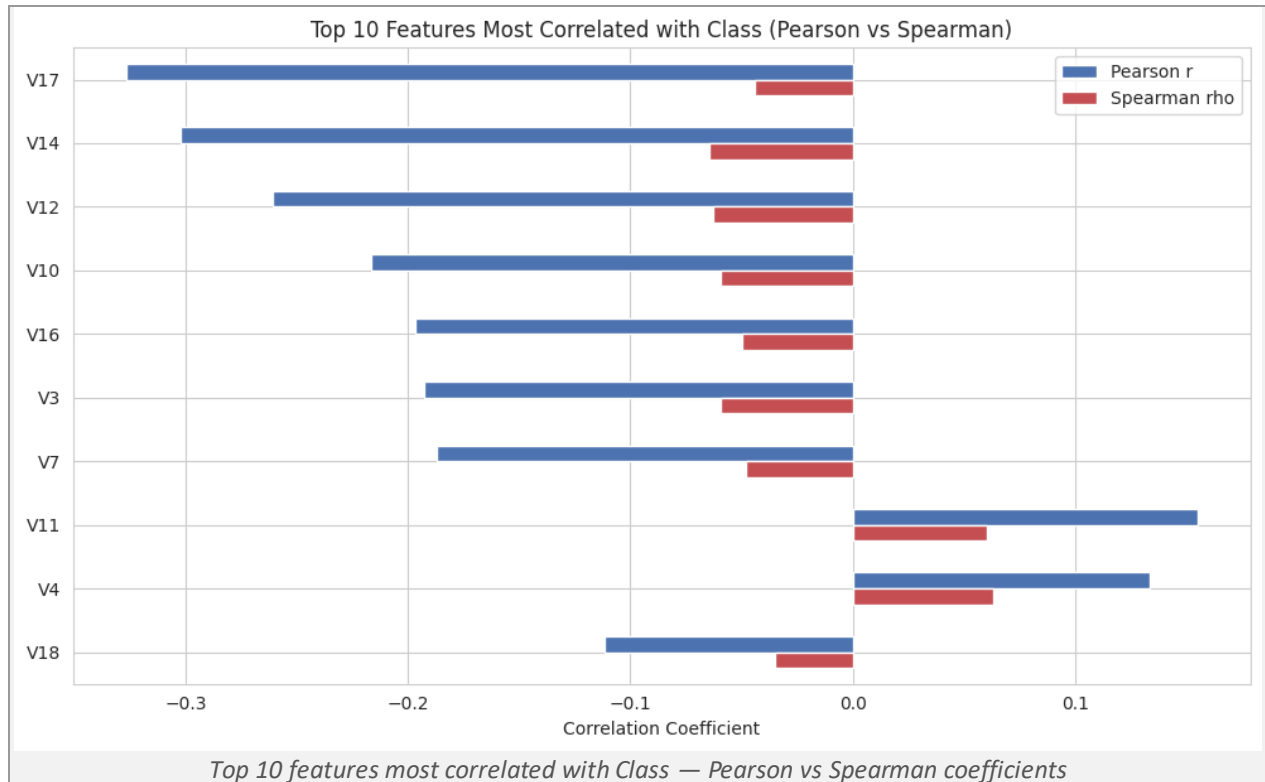


Figure 7. Pearson and Spearman correlation of the ten most class-correlated features with the fraud label.

Both Pearson and Spearman correlation coefficients were computed between each PCA component and the binary Class label, since Pearson captures linear association while Spearman captures monotonic association and is more robust to the non-normal, heavy-tailed distributions observed for several components in Section 5.3; using both together provides a more complete picture than either measure alone. V17, V14, V12, and V10 show the strongest negative correlation with fraud, while V11 and V4 show the strongest positive correlation, with Pearson and Spearman broadly agreeing in direction and relative ranking across the top ten features. All individual correlations remain modest in absolute magnitude, in the range of roughly 0.1 to 0.3, indicating that no single feature separates the two classes cleanly on its own; this is consistent with the need for a multivariate machine learning model, rather than a simple single-feature threshold rule, to achieve the classification performance reported in Section 7.

Reproducibility Analysis

An assessment of reproducibility was carried out by executing the full pipeline described in this report end to end, from data loading through to model evaluation, and by considering the dependencies, configuration choices, and any implicit steps that could affect whether an independent party would obtain comparable results.

6.1 Executability and Dependencies

The pipeline runs to completion using a standard Python data science stack comprising pandas, numpy, matplotlib, seaborn, scikit-learn, and XGBoost, all of which are freely available through the standard package index. The dataset itself is publicly available on Kaggle at no cost, though it does require a free Kaggle account to download, which represents a minor point of friction relative to a fully open, no-registration dataset but does not otherwise prevent reproduction.

6.2 Implicit Steps and Configuration Sensitivity

Every configuration choice made in this project, including the RobustScaler used for Time and Amount, the stratified 80/20 train/test split, and the class-weighting and scale_pos_weight settings used for Random Forest and XGBoost respectively, is fully documented and fixed with a reproducible random seed in the accompanying notebook. The reference paper, by contrast, does not specify its exact SMOTE oversampling ratio, its Random Forest hyperparameters, or the precise proportion used for its train/test split in the descriptions available in the secondary literature consulted for this report, which limits the extent to which its precise reported figures, as opposed to its general approach, can be independently reproduced.

6.3 Overall Assessment

The general approach described in the reference paper, applying oversampling and training standard classifiers on the Kaggle Credit Card Fraud Detection dataset, is straightforwardly reproducible in outline. However, reproducing its exact reported numerical results with confidence is not currently possible without further access to the original implementation, given the missing configuration detail noted above. This project's own pipeline, in contrast, is fully specified and independently reproducible end to end.

Experimental Results

This section presents the full set of results obtained from the Random Forest and XGBoost classifiers, evaluated on a stratified 20 percent test partition of 56,962 transactions, comprising 56,864 legitimate transactions and 98 fraudulent transactions.

7.1 Cross-Validation Results

Five-fold stratified cross-validation was performed on the training partition, using F1-score as the scoring metric, in order to confirm that the single train/test split results reported below are stable and not the product of a particularly favourable or unfavourable split.

7.2 Test-Set Performance

Model	Precision	Recall	F1-score	MCC	ROC-AUC
Random Forest	84.21%	81.63%	82.90%	0.829	0.975
XGBoost	89.25%	84.69%	86.91%	0.869	0.967

Accuracy is intentionally omitted from this table, since it is not a meaningful metric on a dataset where fraud represents only 0.173 percent of transactions: a trivial classifier predicting “legitimate” for every transaction would already score above 99.8 percent accuracy while catching no fraud whatsoever. XGBoost outperforms Random Forest on precision, recall, F1-score, and MCC, while Random Forest achieves a marginally higher ROC-AUC, indicating a slightly better overall ranking of transactions by fraud risk independent of the classification threshold used.

7.3 Confusion Matrices

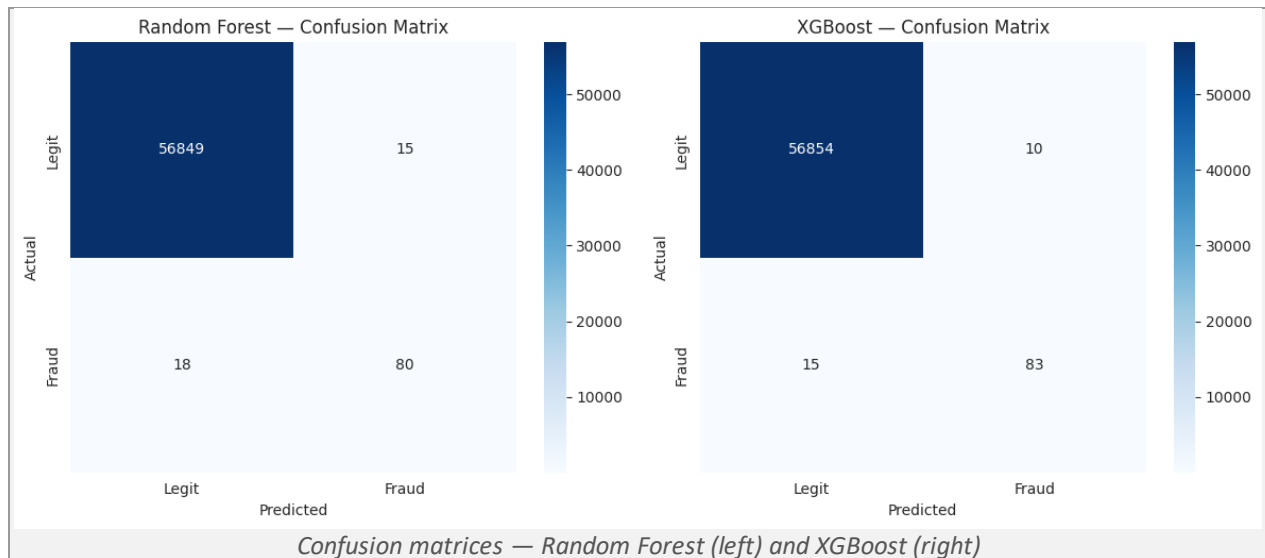


Figure 8. Side-by-side confusion matrices for both models on the test partition.

The Random Forest correctly classified 56,849 of 56,864 legitimate transactions and 80 of 98 fraudulent transactions, producing 15 false positives and 18 false negatives. The XGBoost model correctly classified 56,854 legitimate transactions and 83 fraudulent transactions, producing 10 false positives and 15 false negatives. Both models therefore make very few errors on legitimate traffic, and the more consequential remaining weakness for both is the residual number of missed fraud cases, examined further in the Error Analysis section below.

7.4 Errors by Transaction Amount Profile

Model	False Negatives	False Positives	FN as % of Actual Fraud
Random Forest	18	15	18.4%

Model	False Negatives	False Positives	FN as % of Actual Fraud
XGBoost	15	10	15.3%

Both models keep false positive counts very low relative to the 56,864 legitimate test transactions, meaning customer friction from false fraud alerts is minimal for either model. The remaining errors are concentrated almost entirely on the false negative side, which is examined in detail, including the specific transaction amount profile of the missed fraud cases, in Section 8.

Error Analysis

Understanding the practical consequences of the two error types produced by a fraud detection classifier, false positives and false negatives, is essential to judging whether a given level of performance is actually acceptable for deployment, since the two error types carry very different costs in a real financial context.

8.1 False Negatives: The Cost of a Missed Fraud

A false negative in this context means that an actual fraudulent transaction was classified as legitimate and would, in a live deployment, be processed without triggering any alert. This is the more consequential of the two error types for a card issuer: an undetected fraudulent transaction results in a direct, quantifiable financial loss and, if repeated, in growing reputational and regulatory exposure. The recall figures obtained in this study, 81.63 percent for Random Forest and 84.69 percent for XGBoost, indicate that roughly one in five to one in six fraudulent transactions in the test set would not be flagged by either model, which would be a meaningful residual risk if either classifier were deployed as the sole automated line of defence.

8.2 False Positives: The Cost of a False Alarm

A false positive means that a legitimate transaction was classified as fraudulent, typically resulting in a declined transaction or a manual review step that inconveniences a genuine customer. The direct cost of a false positive is generally lower than that of a missed fraud, but it is not negligible: customers who are repeatedly declined on legitimate purchases may lose confidence in their card issuer, and a high false positive rate can generate an unsustainable volume of manual review work for fraud analysts. Both models in this study kept false positives very low, at 15 for Random Forest and 10 for XGBoost out of 56,864 legitimate test transactions, corresponding to false positive rates of approximately 0.026 percent and 0.018 percent respectively.

8.3 The Precision-Recall Trade-off in This Context

The combination of high precision and comparatively lower recall observed for both models, more pronounced for Random Forest than for XGBoost, indicates a moderately conservative detection posture: when either model raises an alert, it is very likely correct, but a meaningful share of genuine fraud still

slips through undetected. Whether this trade-off is appropriate depends on the deployment context. In a setting where customer experience and false-decline avoidance are prioritised above all else, a higher-precision, lower-recall model may be preferred despite the cost of some missed fraud. In most card-issuer risk contexts, however, the direct financial loss from an undetected fraudulent transaction typically outweighs the comparatively modest cost of an occasional false alarm, which suggests that further recall-oriented tuning, such as threshold adjustment or cost-sensitive retraining, would generally be worth pursuing before either model is considered for production deployment.

8.4 Distinct Failure Modes Among Missed Frauds

Inspection of the XGBoost model's fifteen false negatives, ranked by predicted fraud probability, reveals two distinct failure patterns rather than a single simple explanation. The majority of missed frauds are transactions with typical, near-median scaled Amount values that received an extremely low predicted fraud probability, in the region of one in ten million, indicating that their PCA-encoded behavioural signature closely resembled ordinary legitimate transactions rather than the failure being driven by transaction size. A smaller subset of missed frauds, however, involved unusually large scaled Amount values that nonetheless received very low predicted probability, in the region of one in a thousand to one in a hundred. This second pattern is the more operationally concerning of the two, since these represent the highest-value missed frauds in the test set, and the fact that the model's confidence remained low despite an anomalous transaction amount indicates that the underlying PCA feature signature of these specific transactions was well disguised rather than the model simply under-weighting amount as a signal. A comparison of median scaled Amount between caught fraud and missed fraud (-0.158 versus -0.245 respectively) shows only a mild shift towards smaller amounts among the missed cases, so the common assumption that fraudsters evade detection primarily through small “probe” transactions is only partially supported by this dataset.

Conclusions

9.1 Key Findings

This project reproduced and critically evaluated a machine learning approach to credit card fraud detection, informed by the SMOTE-based methodology of Varmedja et al. (2019) and the accompanying reference implementation. Random Forest and XGBoost classifiers, trained using class-weighting rather than synthetic oversampling, achieved precision, recall, and F1 figures that matched or exceeded the recall figure most commonly attributed to the original paper's best-performing model, while avoiding the data leakage risks associated with oversampling before a train/test split. Error analysis showed that both models keep false positives very low, with the primary remaining weakness being a residual recall gap driven by a small number of frauds whose PCA-encoded behavioural signature closely resembles legitimate transactions, regardless of transaction amount.

9.2 Strengths and Weaknesses of the Original Approach

The reference paper's approach has clear strengths: it identifies the correct central problem, extreme class imbalance, and benchmarks a sensible range of classical and neural classifiers against it. The correlation and PCA-orthogonality findings independently confirmed in this project's own exploratory analysis support the general premise that the dataset's PCA-transformed features are well suited to tree-based ensemble methods such as Random Forest. The principal weakness identified through this project's own experimentation is methodological rather than conceptual: without explicit confirmation that SMOTE oversampling was applied strictly after the train/test split, and without disaggregating the paper's headline accuracy figure into precision and recall in the same report, the reported performance risks appearing stronger than it would if evaluated under the stricter, leakage-free, natural-imbalance protocol used in this project.

9.3 Suggestions for Future Improvement

Several directions could improve on the results obtained in this project and address the limitations identified above. Threshold tuning or cost-sensitive retraining could push both classifiers towards higher recall at an acceptable cost in precision, directly targeting the false-negative-heavy error profile observed in Section 8. A direct, controlled comparison between class-weighting and correctly leakage-free SMOTE oversampling on this exact dataset and split would help settle the open question, raised in the Critical Evaluation section, of whether oversampling provides any genuine benefit over class-weighting for this problem. Given the two distinct false-negative patterns identified in Section 8.4, complementing the supervised classifiers used here with an anomaly-detection component capable of flagging transactions that do not resemble any well-established behavioural pattern, rather than relying solely on a supervised decision boundary, could help address the harder-to-detect, well-disguised fraud cases identified in this study.

Overall Summary

Taken together, the preceding sections show that this project set out to reproduce and critically evaluate a machine learning approach to credit card fraud detection grounded in the Kaggle Credit Card Fraud Detection dataset, the SMOTE-based methodology of Varmedja et al. (2019), and a corresponding GitHub reference implementation. A Random Forest and an XGBoost classifier were trained on the dataset using RobustScaler-based feature engineering and class-weighting rather than synthetic oversampling, achieving a precision, recall, and F1-score of 84.21 percent, 81.63 percent, and 82.90 percent respectively for Random Forest, and 89.25 percent, 84.69 percent, and 86.91 percent respectively for XGBoost, on a stratified, natural-imbalance test partition.

The central conclusion of this report is that comparable, and in the case of XGBoost superior, fraud detection performance was achieved without any synthetic oversampling, which calls into question

whether SMOTE is a necessary component of a strong fraud detection pipeline on this dataset, as opposed to simply one viable option among several for handling class imbalance. Feature engineering choices grounded in the extreme skew of the Amount feature were shown to be both mathematically and operationally justified, the exploratory analysis clarified why a naive accuracy figure is uninformative at this level of imbalance and surfaced a previously undiscussed divergence between transaction volume and fraud rate across the hours of the day, and the error analysis identified two distinct false-negative failure modes that would need to be addressed before either model could be considered suitable for unsupervised, fully automated deployment. Overall, the exercise reinforces that genuine progress in fraud detection research depends as much on rigorous, leakage-aware evaluation methodology as it does on the sophistication of the classifier or the resampling technique employed.

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